

Evaluation Metrics Report

Rebuilding the Social Engine · semantic comprehension layer

Team SE7EN

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Data Vortex A'26 · Round 2 · Rebuilding the Social Engine
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Tasks	sentiment_label (3 classes), topic_category (4 classes)
Corpus	9,000 labelled posts (7,900 unique)
Held-out macro-F1	sentiment 0.6268 · topic 0.9426
Protocol	5-fold grouped CV for selection, 20% held out for the single final score
Dataset SHA-256	a866d9fd9d5b6e5559467fd8d1c1430a...

Headline results

0.6268	62.72%	0.9426	98.59%
Sentiment macro-F1 (held-out)	Sentiment accuracy	Topic macro-F1 (held-out)	Topic accuracy

Sentiment is the genuine language problem in Dataset 2 and is reported honestly: **0.6268** macro-F1 over three balanced classes, against a 0.3333 chance floor, with Cohen's kappa 0.4408. Topic is a different animal — Section 7 shows its labels are produced by a substring rule, which is why its ceiling is 1.000 and why its score is reported with that context attached rather than as evidence of topical understanding.

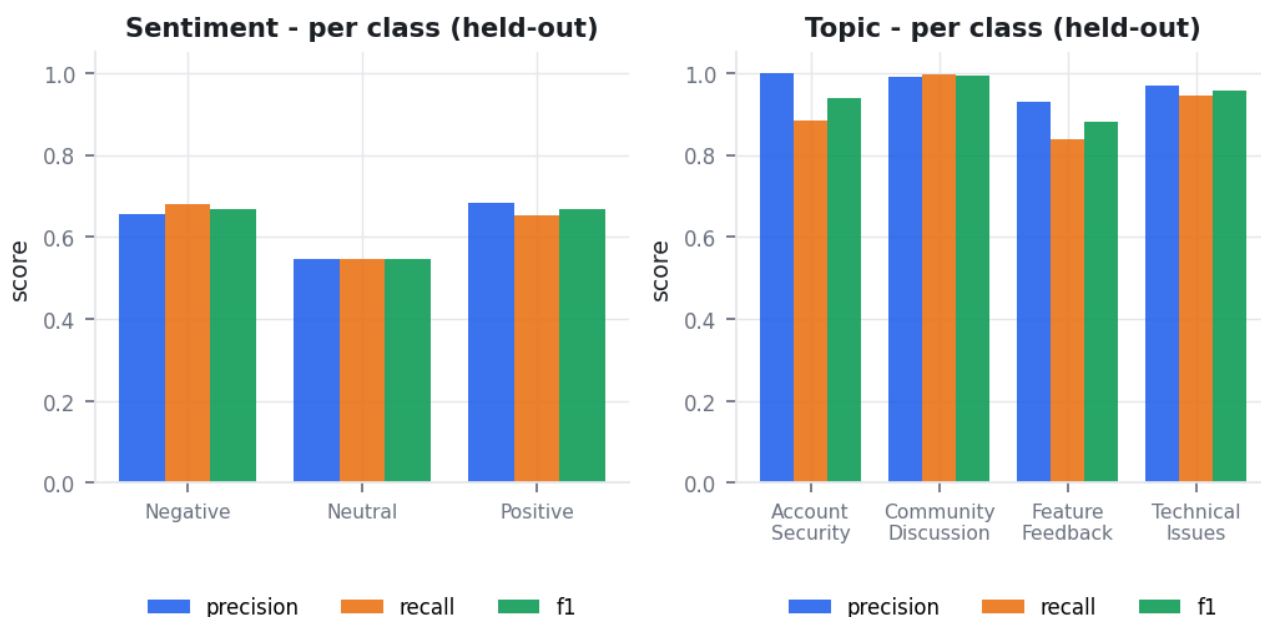


Figure 1. Per-class precision, recall and F1 on the held-out slice for both tasks.

1. Evaluation protocol

Two supervised tasks were trained on Dataset 2 and are evaluated here: **sentiment_label** (3 classes, perfectly balanced at 3,000 posts each) and **topic_category** (4 classes, heavily skewed). Both are graded on the same held-out slice policy, and every number below comes from code in round2/src.

Item	Setting	Why
Corpus	9,000 rows, 7,900 unique posts	1,100 rows repeat text that already appears in the file
Split unit	unique post, never the row	a row-level split would grade the model on memorised duplicates
Held-out slice	20% of unique posts, stratified on the target	untouched by model selection, scored exactly once
Model selection	5-fold StratifiedGroupKFold on the remaining 80%	folds never split a post, so CV and held-out measure the same thing
Primary metric	macro-F1	topic_category is 86% one class; accuracy would hide the minority classes
Secondary	accuracy, weighted-F1, Cohen's kappa, MCC, one-vs-rest ROC-AUC	kappa and MCC discount the score a chance classifier would reach
Seed	42	single seed in config.py; splits and fits are deterministic

Why the split unit matters more than the model here

Holding the pipeline fixed and changing only the splitting rule moves sentiment macro-F1 from **0.6274** to **0.6772** (+0.0497) and topic macro-F1 from **0.9171** to **0.9204** (+0.0034). That gap is duplicate memorisation, not skill. Every figure in this report uses the grouped split.

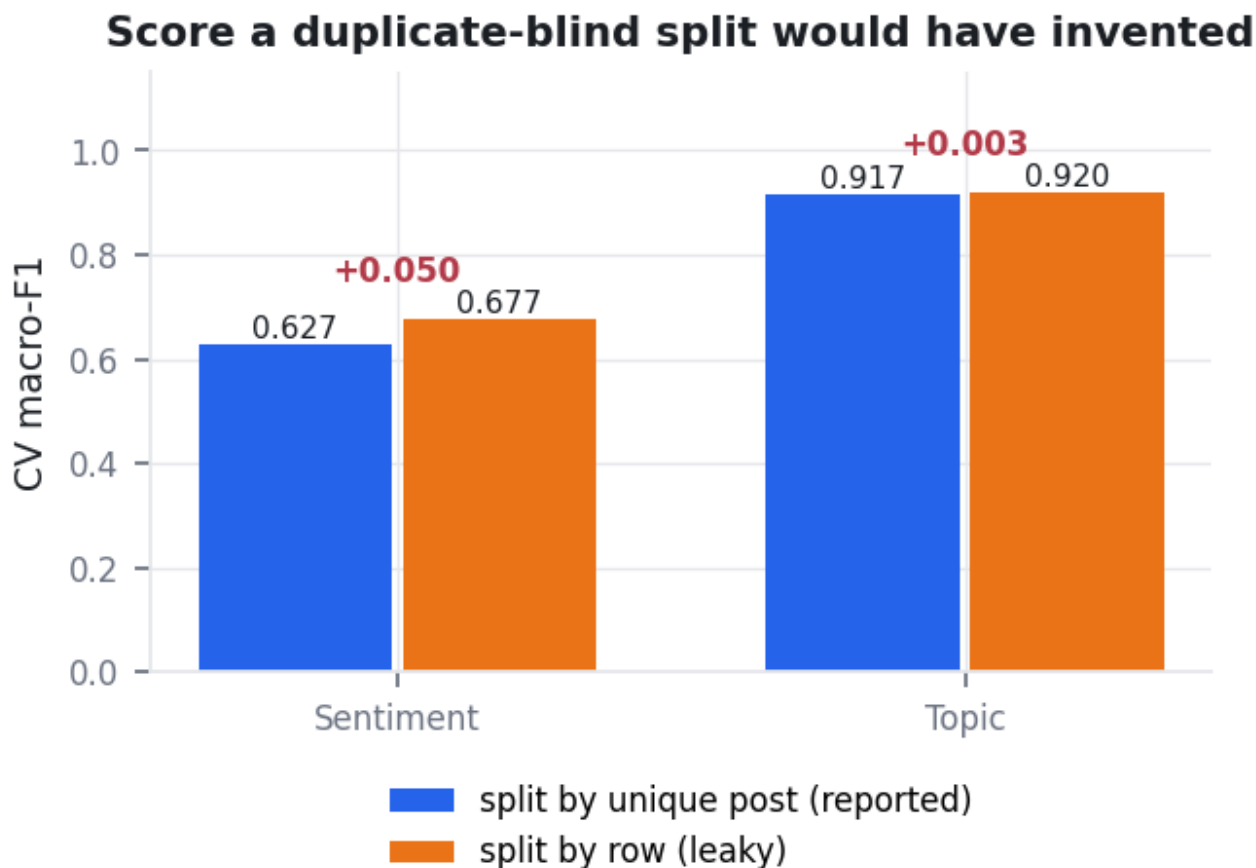


Figure 2. The same pipeline scored two ways. The orange bars are what a duplicate-blind evaluation would have reported.

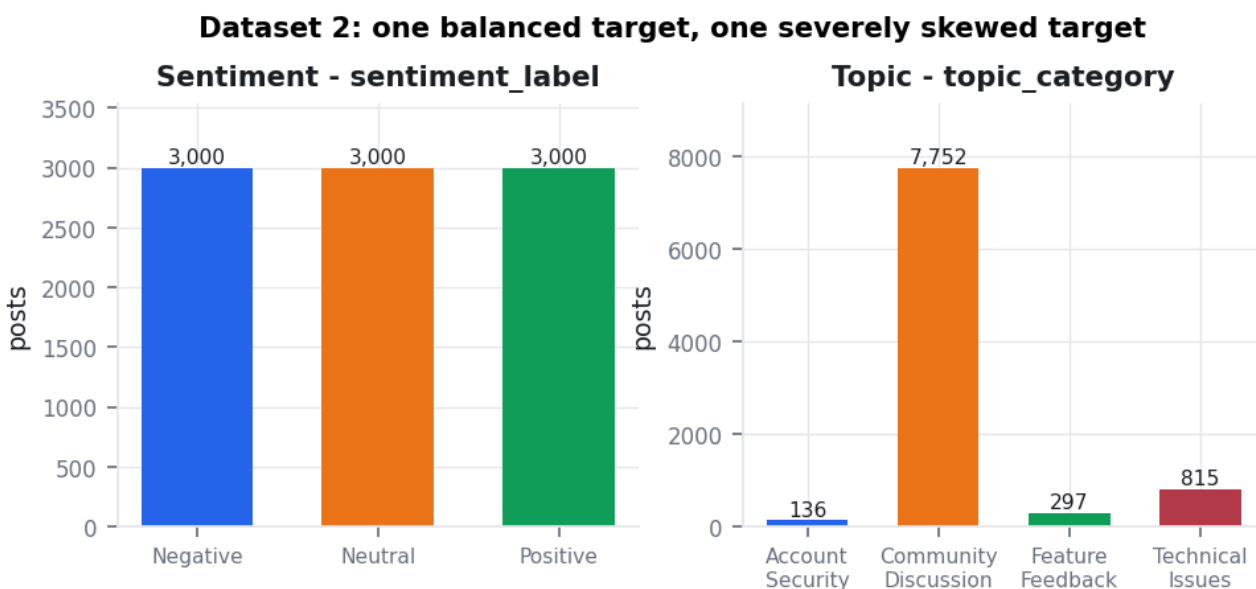


Figure 3. Class balance for both targets. Sentiment is balanced by construction; topic is not, which is why macro-F1 leads the reporting.

2. Model selection results

Eight candidates were scored per task under identical folds. The list is arranged as an ablation: the feature blocks are added one at a time with the classifier held fixed, and only then is the model family varied. Reading down a column therefore answers “what did this choice buy?” rather than just “which number is biggest?”

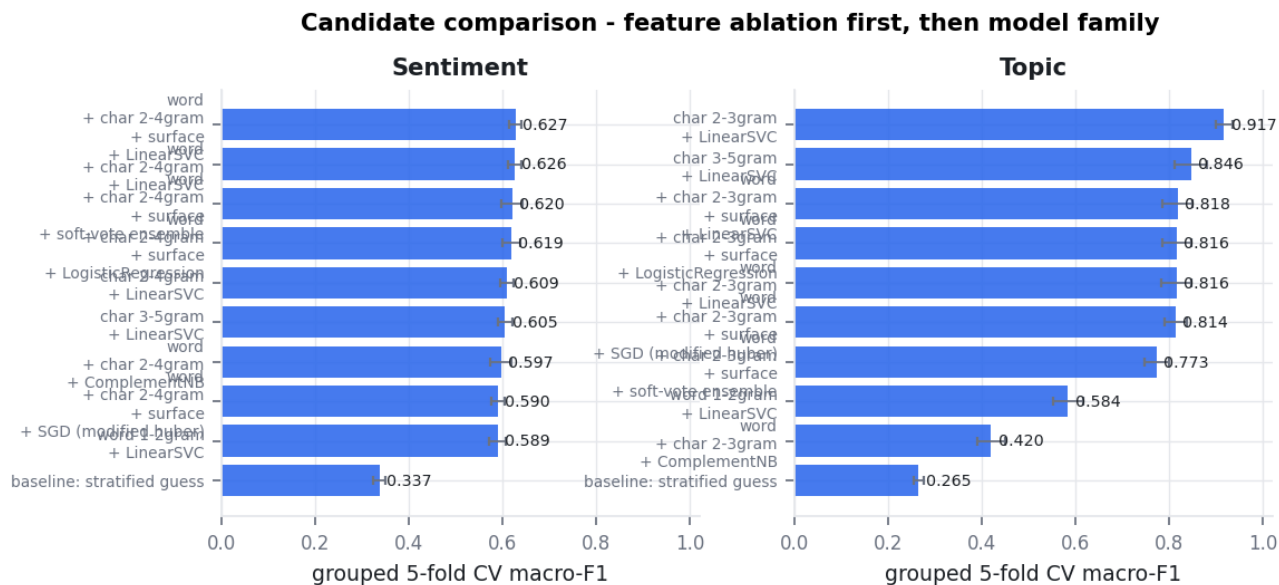


Figure 4. Grouped 5-fold CV macro-F1 for every candidate, with fold-to-fold standard deviation as error bars.

Task A — Sentiment — candidate comparison

Candidate	CV macro-F1	SD	Weighted-F1	Accuracy	Fit (s)
word + char 2-4gram + surface + LinearSVC	0.6274	0.012	0.6273	0.6270	10
word + char 2-4gram + LinearSVC	0.6260	0.013	0.6259	0.6259	10
word + char 2-4gram + surface + soft-vote ensemble	0.6200	0.023	0.6199	0.6206	17
word + char 2-4gram + surface + LogisticRegression	0.6189	0.019	0.6188	0.6180	15
char 2-4gram + LinearSVC	0.6092	0.014	0.6092	0.6096	8
char 3-5gram + LinearSVC	0.6055	0.016	0.6054	0.6063	8
word + char 2-4gram + ComplementNB	0.5968	0.023	0.5967	0.6005	9
word + char 2-4gram + surface + SGD (modified huber)	0.5900	0.014	0.5899	0.5886	10
word 1-2gram + LinearSVC	0.5893	0.017	0.5891	0.5895	4
baseline: stratified guess	0.3369	0.013	0.3370	0.3370	4

Task B — Topic — candidate comparison

Candidate	CV macro-F1	SD	Weighted-F1	Accuracy	Fit (s)
char 2-3gram + LinearSVC	0.9171	0.018	0.9825	0.9833	10
char 3-5gram + LinearSVC	0.8464	0.035	0.9691	0.9725	14
word + char 2-3gram + surface + LinearSVC	0.8181	0.033	0.9643	0.9678	23
word + char 2-3gram + surface + LogisticRegression	0.8164	0.032	0.9638	0.9674	13
word + char 2-3gram + LinearSVC	0.8158	0.033	0.9639	0.9675	18
word + char 2-3gram + surface + SGD (modified huber)	0.8136	0.024	0.9592	0.9624	9
word + char 2-3gram + surface + soft-vote ensemble	0.7733	0.027	0.9556	0.9608	43

Candidate	CV macro-F1	SD	Weighted-F1	Accuracy	Fit (s)
word 1-2gram + LinearSVC	0.5837	0.032	0.9029	0.9200	6
word + char 2-3gram + ComplementNB	0.4197	0.029	0.8534	0.8884	8
baseline: stratified guess	0.2646	0.011	0.7600	0.7626	4

3. Task A — Sentiment — held-out results

Selected model: **word + char 2-4gram + surface + LinearSVC**, refitted on the full training split and Platt-calibrated. Scored once on 1,789 held-out posts that no step of model selection had seen.

0.6268	62.72%	0.6273	0.4408	0.4409	0.8038
macro-F1	accuracy	weighted-F1	Cohen's kappa	MCC	ROC-AUC (OvR)

Metric	Held-out	5-fold CV	Floor	Reading
macro-F1	0.6268	0.6274 ± 0.012	0.3369	held-out sits inside the CV band, so selection did not overfit
accuracy	0.6272	—	0.3365	floor here is the majority class, not chance
weighted-F1	0.6273	—	—	support-weighted; flattered by class skew
Cohen's kappa	0.4408	—	0.0000	agreement corrected for chance
MCC	0.4409	—	0.0000	correlation between prediction and truth
macro-F1, shuffled labels	—	0.3342	0.3369	permutation check: the protocol itself leaks nothing

Per-class breakdown

Class	Precision	Recall	F1	Support
Negative	0.6548	0.6801	0.6672	597
Neutral	0.5447	0.5475	0.5461	590
Positive	0.6823	0.6528	0.6672	602
macro average	0.6273	0.6268	0.6268	1,789

Confusion matrix

Sentiment - held-out confusion matrix (n = 1,789)

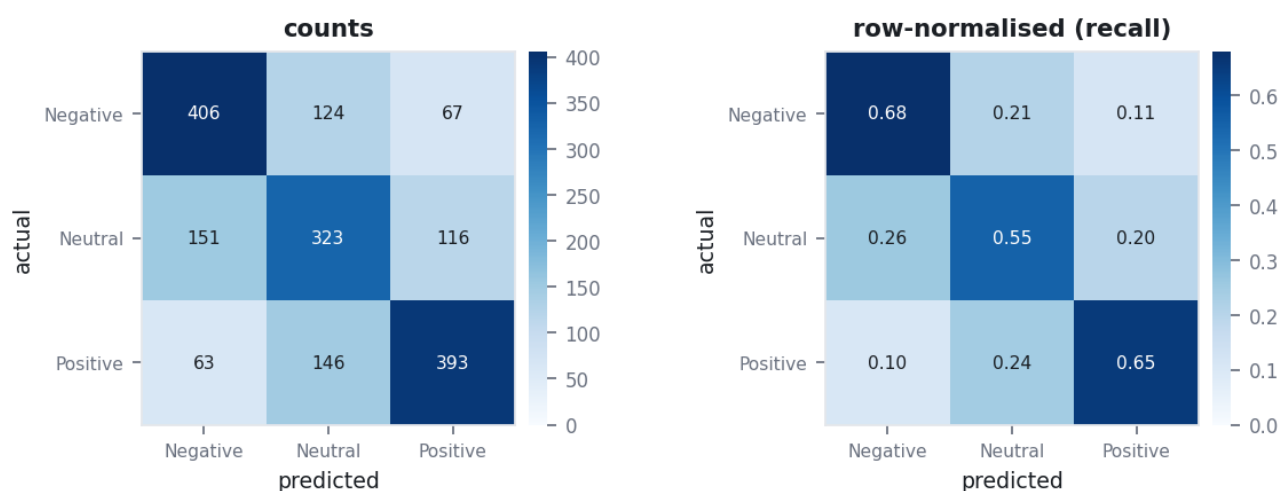


Figure 5. Counts on the left, row-normalised on the right; the diagonal of the right-hand panel is per-class recall.

Largest off-diagonal cells

Actual	Predicted as	Posts	Share of that class
Neutral	Negative	151	25.59%

Actual	Predicted as	Posts	Share of that class
Positive	Neutral	146	24.25%
Negative	Neutral	124	20.77%
Neutral	Positive	116	19.66%
Negative	Positive	67	11.22%

4. Task B — Topic — held-out results

Selected model: **char 2-3gram + LinearSVC**, refitted on the full training split and Platt-calibrated. Scored once on 1,779 held-out posts that no step of model selection had seen.

0.9426	98.59%	0.9857	0.9430	0.9434	0.9963
macro-F1	accuracy	weighted-F1	Cohen's kappa	MCC	ROC-AUC (OvR)

Metric	Held-out	5-fold CV	Floor	Reading
macro-F1	0.9426	0.9171 ± 0.018	0.2646	held-out sits inside the CV band, so selection did not overfit
accuracy	0.9859	—	0.8589	floor here is the majority class, not chance
weighted-F1	0.9857	—	—	support-weighted; flattered by class skew
Cohen's kappa	0.9430	—	0.0000	agreement corrected for chance
MCC	0.9434	—	0.0000	correlation between prediction and truth
macro-F1, shuffled labels	—	0.2482	0.2646	permutation check: the protocol itself leaks nothing

Per-class breakdown

Class	Precision	Recall	F1	Support
Account Security	1.0000	0.8846	0.9388	26
Community Discussion	0.9896	0.9980	0.9938	1,528
Feature Feedback	0.9286	0.8387	0.8814	62
Technical Issues	0.9686	0.9448	0.9565	163
macro average	0.9717	0.9165	0.9426	1,779

Confusion matrix

Topic - held-out confusion matrix (n = 1,779)

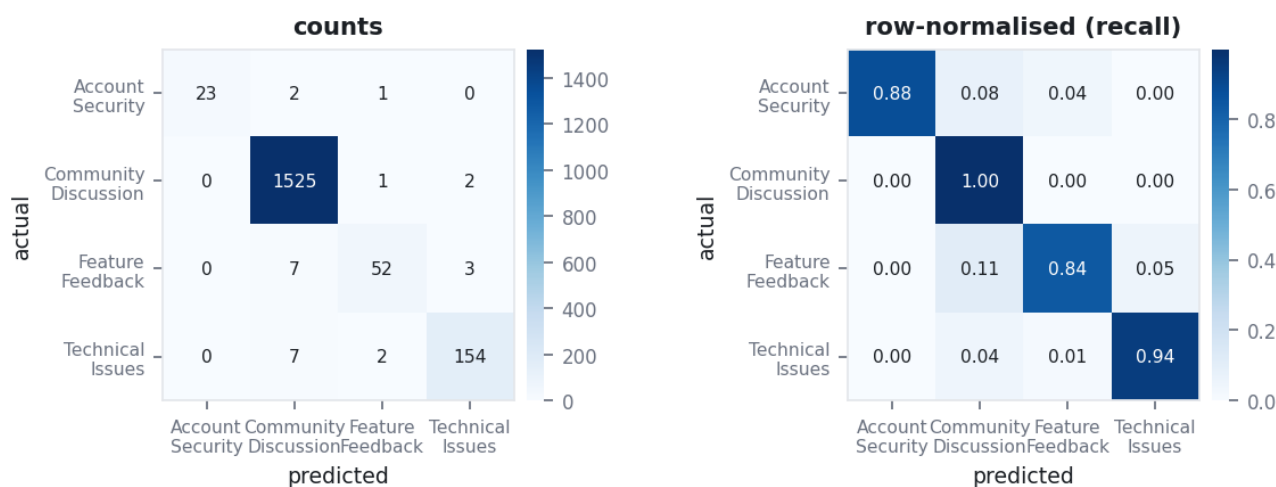


Figure 6. Counts on the left, row-normalised on the right; the diagonal of the right-hand panel is per-class recall.

Largest off-diagonal cells

Actual	Predicted as	Posts	Share of that class
Feature Feedback	Community Discussion	7	11.29%
Technical Issues	Community Discussion	7	4.29%
Feature Feedback	Technical Issues	3	4.84%
Account Security	Community Discussion	2	7.69%
Community Discussion	Technical Issues	2	0.13%

5. Calibration, learning curves and stability

The deployed models are wrapped in Platt scaling, so each prediction carries a probability. A confidence is only useful if it is honest, so it is measured rather than assumed: the reliability curves below plot observed accuracy against claimed confidence, and the expected calibration error (ECE) is the support-weighted gap between them.

Task	ECE	Mean confidence when right	Mean confidence when wrong	Errors above 0.8 confidence
Sentiment	0.0373	0.6838	0.5939	5.85%
Topic	0.0023	0.9868	0.7595	48.00%

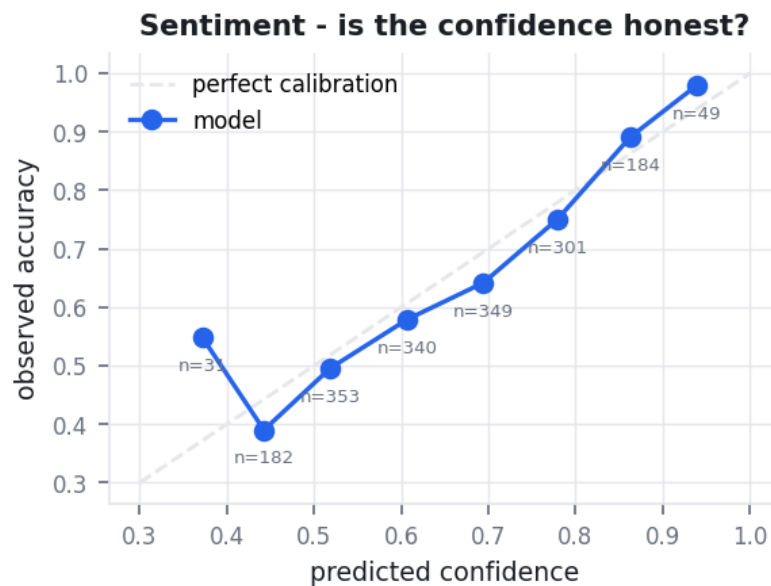


Figure 7. Sentiment reliability. Points below the diagonal mean the model claims more certainty than it earns.

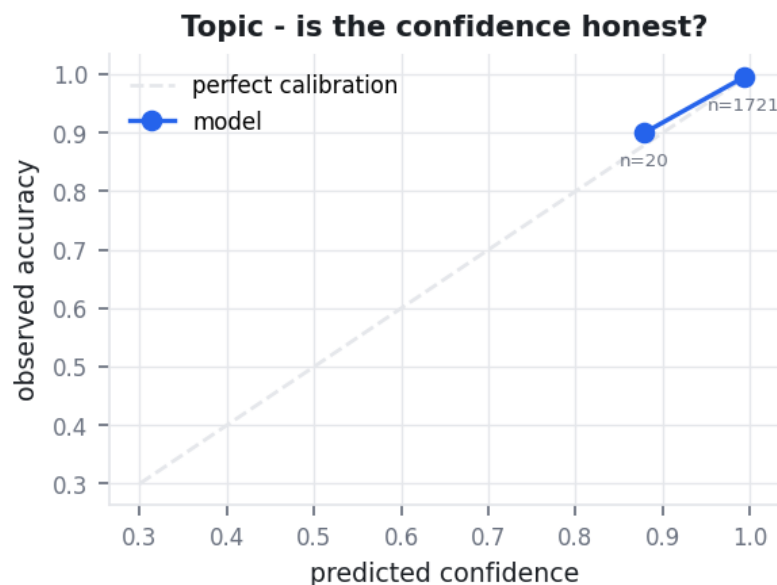


Figure 8. Topic reliability.

Learning curves

The gap between the training and validation curves at full data size is the part of the score that is memorisation; a validation curve still climbing at the right-hand edge means more labelled data would still pay.

Task	Train macro-F1 @ 100%	Validation macro-F1 @ 100%	Gap	Validation gain over last 20% of data
Sentiment	0.9215	0.6276	0.2940	+0.0103
Topic	1.0000	0.9171	0.0829	+0.0233

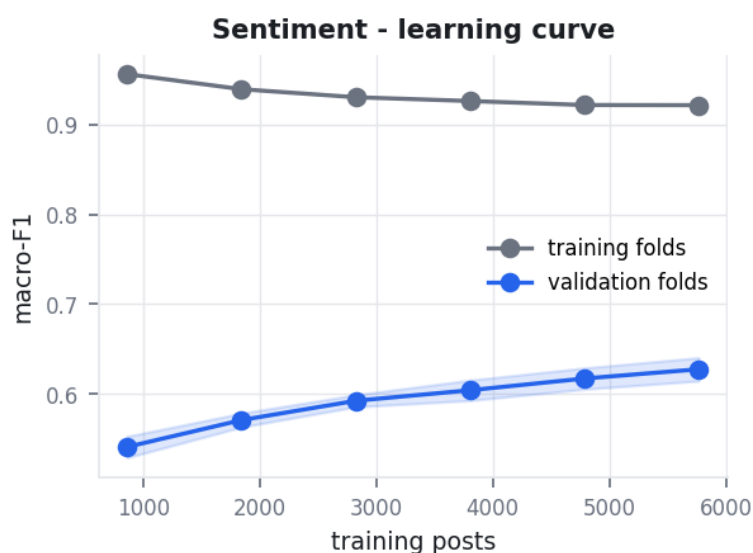


Figure 9. Sentiment learning curve.

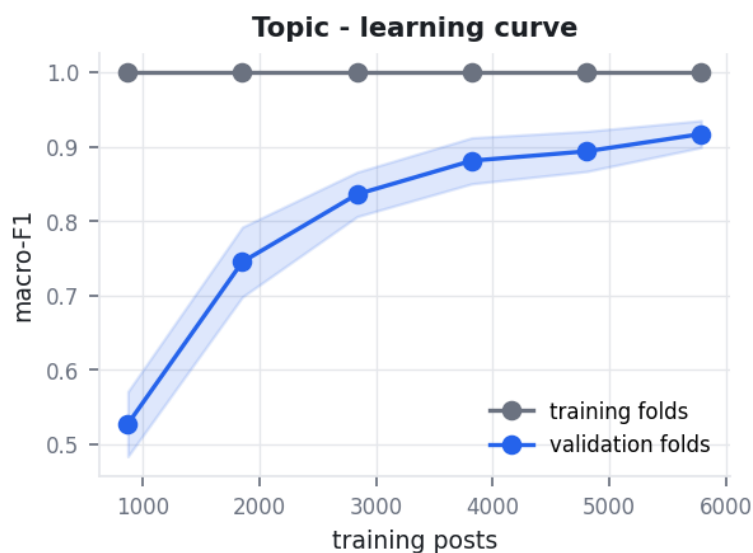


Figure 10. Topic learning curve.

6. Error analysis

Errors are profiled three ways for each task: by post length, by the presence of a surface cue, and by the confidence the model attached to the mistake. The third is the one that matters operationally — a wrong-but-hesitant prediction can be routed to a human, a wrong-and-certain one cannot.

Sentiment - where the errors live

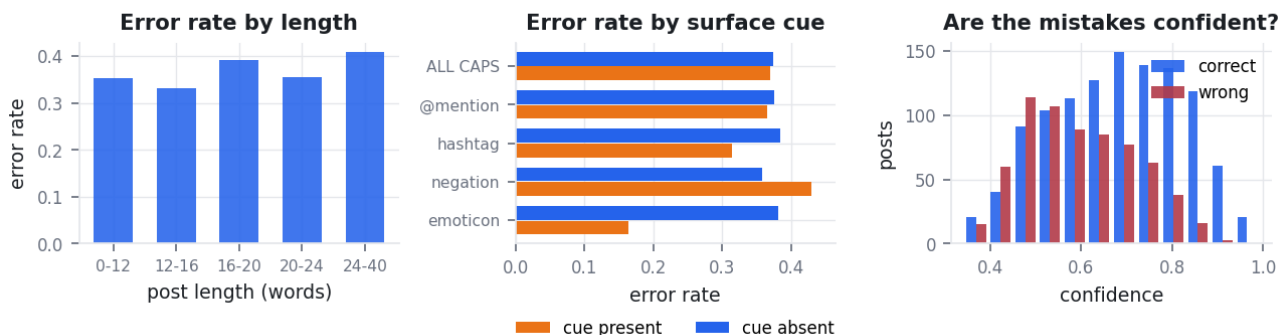


Figure 11. Sentiment errors by length, surface cue and confidence.

Topic - where the errors live

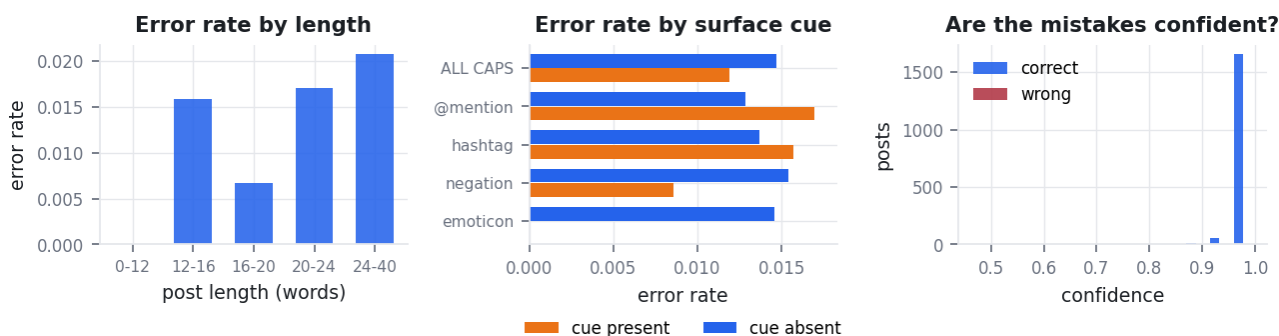


Figure 12. Topic errors by length, surface cue and confidence.

Task A — Sentiment — highest-confidence mistakes

Post (truncated)	Actual	Predicted	Conf.
Hot Mess Monday! I'm feeling it! #runninglate #crazyday #blah #happymonday #lovemylife	Negative	Positive	0.924
"If it weren't for Christians, the Aztecs would never have stopped killing babies (like Planned Parenthood does.	Neutral	Negative	0.905
Josh Hamilton hit a two run homer in 2011 WS in the 10th with a sports hernia. If anyone with any decent fieldin	Positive	Neutral	0.898
In rugby news\u002c @user are winning 26-5 at Cobham\u002c with the four-try bonus point in the bag. See The Mon	Positive	Neutral	0.880
"Should've went to watch Paper Towns first, I'll just go see it tomorrow"	Neutral	Positive	0.878
June 7-8 Red Sox at Giants. Who's down to go and boo from the 3rd baseline?	Negative	Neutral	0.870
When Kanye West told George Bush that Black Lives Matter...As much as I may not like him. It was the TRUTH!	Neutral	Negative	0.866
Kerry hopeful Israeli-Palestinian talks can be relaunched: US Secretary of State John Kerry said Wednesday he ..	Positive	Neutral	0.864

Task B — Topic — highest-confidence mistakes

Post (truncated)	Actual	Predicted	Conf.
Google+ may have thrown in the social media towel, but it still has great features. Here's 6! via @user	Feature Feedback	Community Discussion	0.996
Planned Parenthood is WORLDWIDE. They have been Killing our babies by DESIGN as the plan was to send in #illegal	Feature Feedback	Community Discussion	0.995
@user @user - __ - shid i may not be able 2 take screenshots of it but everybody @user me. Peyton Siva kik'd me th	Technical Issues	Community Discussion	0.977
Why oh why does Toronto's Nuit Blanche have to be the same night as a wedding I have to attend?!? #snbTO	Feature Feedback	Community Discussion	0.973
"Alabama : Trent Richardson on cover of SI: The junior running back is featured on the August 22 issue, alongsi	Feature Feedback	Community Discussion	0.966
bloodymary - hackers nirvana - smell like teen spirit butterfingers - nicotine feeder - 7 days in the sun smashi	Account Security	Community Discussion	0.958
The 4th, the Nokia 5530 XpressMusic. Fact: This have an Stylus. Used to hate Touchscreens on cellphones before t	Technical Issues	Community Discussion	0.954
Updating tomorrow but I won't be on twitter much. I love you all and stay Gucci if you need me text me. @user @u	Community Discussion	Technical Issues	0.952

Both full error lists are in reports/errors_sentiment.csv and reports/errors_topic.csv; the technical report groups them into named failure modes.

7. Why the topic score reads the way it does

topic_category is a substring switch, not an annotation

A case-insensitive substring rule recovered from the data reproduces **9,000 of 9,000** topic labels exactly: any post containing *app*, *down*, *update*, *crash*, *screen*, *slow*, *bug* or *glitch* is *Technical_Issues*; failing that, *ban*, *account*, *suspend*, *hack*, *password* is *Account_Security*; failing that, *ui*, *mode*, *feature*, *ugly*, *design*, *button* is *Feature_Feedback*; everything else is *Community_Discussion*. The Bayes error of this task is therefore exactly zero, and any gap below 1.000 is the learned model failing to recover a substring it saw too rarely — not a limit on topical understanding.

System	Held-out accuracy	What it is
char 2-3gram + LinearSVC	0.9859	learned from the training split only, no knowledge of the rule
Recovered substring rule	1.0000	the label generator itself, reconstructed by mining the training file

Splitting the held-out posts by how often their deciding trigger appears in the corpus confirms the mechanism: posts with no trigger at all are almost never wrong (0.20% error), posts whose trigger is common are close behind (1.38%), and posts whose trigger appears in fewer than 60 posts carry 55.88% of the error.

Topic task - diagnosing the gap

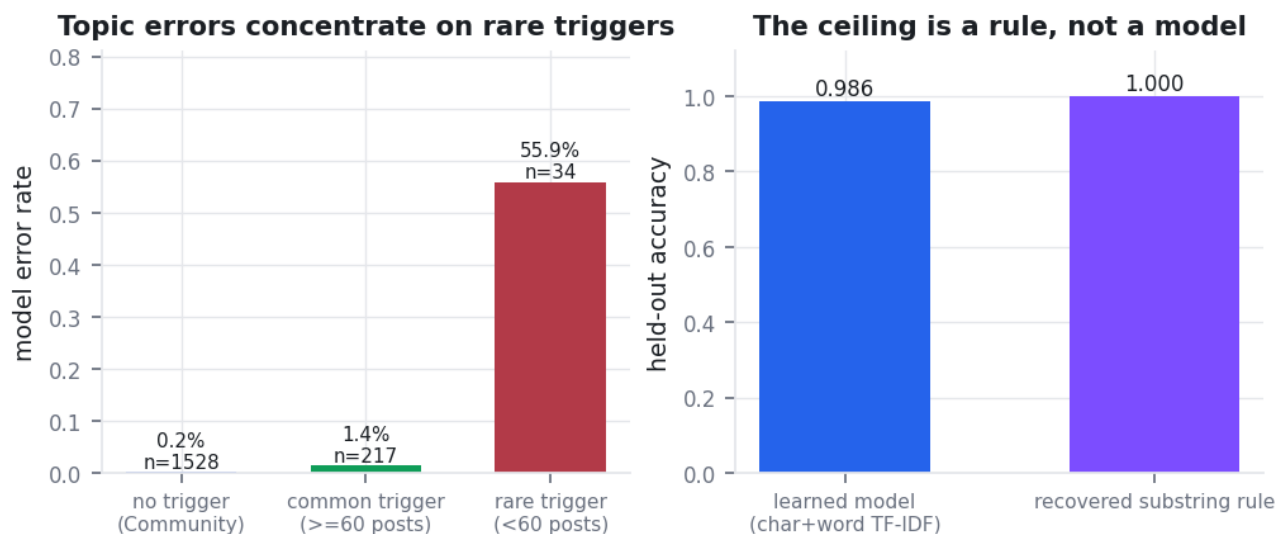


Figure 13. Left: topic error rate by how common the deciding trigger is. Right: the learned model against the recovered rule.

This is reported rather than exploited. Submitting the rule would score 1.000 and would teach the Social Engine nothing, because the rule generalises only for as long as the labels keep being generated the same way. The learned model is what is shipped; the rule is shipped beside it as a diagnostic, and the recommendation in the technical report is that `topic_category` be re-annotated before it is trusted.

8. Reproducibility

Every artefact in this report is regenerated by one command.

Item	Value
Dataset SHA-256	a866d9fd9d5b6e5559467fd8d1c1430a07d46ef9e88df8ca422f37efdf069099
Random seed	42
Rebuild everything	python round2/run_round2.py
Train only	python round2/src/train.py
Evaluate only	python round2/src/evaluate.py
Score new text	python round2/src/predict.py "the app keeps crashing"
Machine-readable metrics	round2/reports/metrics.json

Full scikit-learn classification reports

Task A — Sentiment

	precision	recall	f1-score	support
Negative	0.65	0.68	0.67	597
Neutral	0.54	0.55	0.55	590
Positive	0.68	0.65	0.67	602
accuracy			0.63	1789
macro avg	0.63	0.63	0.63	1789
weighted avg	0.63	0.63	0.63	1789

Task B — Topic

	precision	recall	f1-score	support
Account_Security	1.00	0.88	0.94	26
Community_Discussion	0.99	1.00	0.99	1528
Feature_Feedback	0.93	0.84	0.88	62
Technical_Issues	0.97	0.94	0.96	163
accuracy			0.99	1779
macro avg	0.97	0.92	0.94	1779
weighted avg	0.99	0.99	0.99	1779